

Fraud Shield: Intelligent Risk Scoring & Explainable AI

Credit Card Fraud Detection using Cost-Sensitive ML, Tree SHAP & Dynamic Risk Tiering

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5	0.8966	92.59%	66 / 66
MODELS EVALUATED	TOP COMPOSITE SCORE	TEST PRECISION	TESTS PASSING

Phases 2 & 3 Engineering Milestones

PHASE 2	Cost-Sensitive Model Training: Trained Dummy Baseline, Logistic Regression, Decision Tree, Random Forest, and XGBoost on 170,235 transactions using balanced class weights and asymmetric scale factor (<code>scale_pos_weight = 598.42</code>).
PHASE 2	Threshold Tuning & Zero-Leakage Evaluation: Tuned operating thresholds over validation split maximizing F2-Score . Selected XGBoost and evaluated on 56,746 held-out test transactions: caught 75/95 frauds with only 6 false alarms (99.99% Specificity).
PHASE 3	Calibrated Risk Engine: Engineered 0–100 integer risk score and mapped transactions to operational action categories: LOW (ALLOW) , MEDIUM (MONITOR) , and HIGH (REVIEW) .
PHASE 3	Mathematical Tree SHAP Attribution: Exact additivity formulation in margin space ($\sigma(\text{base} + \Sigma\psi) \equiv P(\text{fraud})$ with error $< 10^{-10}$). Generates local waterfall charts, global importance, and formal non-causal regulatory disclaimers.

Raw Transaction → Feature Engineering (33 Feats) → Scaler Pipeline → XGBoost Inference → Risk Engine (0-100) → SHAP Waterfall Plot

Cost-Sensitive Training & Imbalance Mitigation

Combating extreme 0.172% fraud class imbalance without synthetic distortion

The High Cost of Accuracy in Fraud Detection

In our dataset of 284,807 transactions, fraud occurs in only **0.172%** of cases (1 fraud per 577 legitimate transactions). A naive model unconditionally predicting "Legitimate" achieves **99.83% accuracy** while catching exactly **0% of fraud**. Hence, accuracy is disqualified as a primary metric.

5 Candidate Model Families

- Dummy Baseline:** Prior-stratified coin flip (establishes empirical floor).
- Logistic Regression:** L2-regularized linear baseline ($C=1.0$, `balanced`).
- Decision Tree:** Single rule-tree benchmark (`depth=10`, `balanced`).
- Random Forest:** Bagged ensemble of 200 trees (`depth=15`, `balanced`).
- XGBoost:** Gradient boosted trees with asymmetric penalization (200 trees, `depth=6`).

Mathematical Weighting Strategy

Scikit-learn balanced weights penalize fraud errors by class ratio:

$$w_1 = N / (2 \cdot N_1) \approx 299.71 \times w_0$$

For XGBoost, positive gradient loss is amplified directly by training sample ratio:

$$\text{scale_pos_weight} = N_{\text{legit}} / N_{\text{fraud}} = 169,951 / 284 = 598.42$$

Model Selection Formula & Safety Gates

Operating threshold swept across [0.05, 0.95] in 100 steps on validation split to maximize:

$$\text{Composite Score} = 0.35 \times F2 + 0.30 \times \text{PR-AUC} + 0.20 \times \text{ROC-AUC} + 0.15 \times F1$$

Gate 1: Recall $\geq 70.0\%$

Gate 2: Precision $\geq 10.0\%$

Gate 3: ROC-AUC $\geq 90.0\%$

Validation Model Comparison & Winner Selection

Empirical results across 56,745 validation transactions (94 actual fraud cases)

RANK	CANDIDATE MODEL	THRESH	RECALL	PRECISION	F1	F2	PR-AUC	ROC	COMPOSITE	STATUS
#1	XGBoost (Selected)	0.2773	86.17%	91.01%	0.8852	0.8710	0.8811	0.9734	0.8966	WINNER
#2	Random Forest	0.4864	86.17%	84.38%	0.8526	0.8581	0.8383	0.9689	0.8735	PASSED
#3	Logistic Regression	0.9500	88.30%	33.74%	0.4882	0.6672	0.7882	0.9828	0.7398	PASSED
#4	Decision Tree	0.0500	82.98%	18.18%	0.2983	0.4845	0.4955	0.9139	0.5458	PASSED
#5	Dummy Baseline	0.5000	0.00%	0.00%	0.0000	0.0000	0.0017	0.4992	0.1003	FAILED

XGBoost Confusion Matrix (Thresh = 0.2773)

Pred Legit Pred Fraud Actual Legit 56,643 (TN) 8 (FP) ← Only 8 false alarms! Actual Fraud 13 (FN) 81 (TP) ← 86.17% caught Precision: 91.01% | Specificity: 99.986%

Random Forest vs. Logistic Reg Confusion

Random Forest: 81 TP | 13 FN | 15 FP (84.38% Prec) Logistic Regression: 83 TP | 11 FN | 163 FP (33.74% Prec) Decision Tree: 78 TP | 16 FN | 351 FP (18.18% Prec)

Winning Model Decision Rationale

- 1. Minimal Friction:** Only 8 false alarms across 56,651 customers (nearly half the false alarms of Random Forest).
- 2. Highest PR-AUC (0.8811):** Superior precision-recall curve area unaffected by massive negative class volume.
- 3. Asymmetric Generalization:** Feature/sample subsampling mitigated overfitting to sparse fraud noise.

Held-Out Test Set Performance (Zero Leakage)

Evaluating XGBoost on 56,746 completely unseen transactions (95 fraud cases)

92.59%

TEST PRECISION

78.95%

TEST RECALL (75/95)

99.99%

SPECIFICITY

0.8178

TEST PR-AUC

Held-Out Test Set Confusion Matrix

Pred Legit Pred Fraud Actual Legit 56,645 (TN)
6 (FP) Actual Fraud 20 (FN) 75 (TP) -----
----- • True
Positives (Frauds Stopped): 75 • False
Positives (False Alarms): 6 • False Positive
Rate: 0.0106% • 1 false alarm per 9,440
cardholder purchases!

Strict Anti-Leakage Protocol

- **Zero Test-Set Tuning:** Operating threshold was fixed strictly at 0.2773 from validation search.
- **Pipeline Immutability:** StandardScaler learned mean/std exclusively on training data.
- **High Precision Stability:** Test precision rose from 91.01% (val) to **92.59%** on unseen transactions.

Operational Threshold Sensitivity Trade-offs

THRESHOLD	PRECISION	RECALL	F1-SCORE	F2-SCORE	OPERATIONAL BANKING APPLICATION
0.05	71.93%	87.23%	0.7885	0.8367	High-risk emergency mode / active coordinated attack
0.10	81.00%	86.17%	0.8351	0.8508	Aggressive automated fraud interception
0.2773 (Opt)	91.01%	86.17%	0.8852	0.8710	Fraud Shield Production Default (Max F2)
0.50	93.02%	85.11%	0.8889	0.8658	Standard uncalibrated ML threshold
0.90	94.05%	84.04%	0.8876	0.8587	VIP Cardholder Mode / zero customer friction

Calibrated Risk Engine & Tree SHAP Attribution

Translating raw probabilities into auditable risk scores and mathematical explanations



Tree SHAP: Game-Theoretic Mathematical Additivity

Using Shapley Additive Explanations (Tree SHAP), features act as cooperative players contributing marginal payouts to shift model prediction away from base expectation:

$$\text{Margin}(x) = \psi_0 + \sum \psi_i(x) \implies P(\text{fraud}) = \sigma(\psi_0 + \sum \psi_i(x))$$

$\varphi_i > 0$ (Increases Fraud Risk): Pushes logit margin positive → strictly increases fraud probability.

$\varphi_i < 0$ (Mitigating Evidence): Pushes logit margin negative → supports transaction legitimacy.

✓ Verified in Automated Tests: Numerical difference $|P(\text{fraud}) - \sigma(\text{Margin})| < 10^{-10}$ across all transactions!

Regulatory Compliance: Attribution vs. Causality

Under GDPR Article 22 and FCRA compliance guidelines, machine learning explanations must not imply false human causation. Every explanation generated by Fraud Shield attaches an automated scientific disclaimer:

"Feature contributions indicate statistical associations learned from historical training data. They represent empirical risk signals and do NOT imply real-world causation, motive, or legal culpability."

Global Importance, Case Walkthrough & QA

End-to-end verified inference from raw transaction input to visual explanation

Top Global Fraud Drivers (Mean |SHAP|)

RANK	FEATURE	MEAN SHAP	GLOBAL SHARE	LATENT DOMAIN
#1	V14	2.6514	16.0%	Credential compromise
#2	V4	2.2123	13.3%	Transaction velocity
#3	V3	1.0718	6.5%	Account consistency
#4	V11	0.9787	5.9%	Behavioral anomaly
#5	V12	0.8836	5.3%	Terminal/geo anomaly

Case Walkthrough (Actual Fraud)

Prob: 1.0000 | Score: 100/100 Level: HIGH | Alert: FRAUD ALERT Top Risk Push Factors: 1. V14 (val=-4.58, SHAP=+3.79, 29.6%) 2. V12 (val=-5.31, SHAP=+1.49, 11.6%) 3. Amount (val=0.05, SHAP=+1.42, 11.1%) 4. V10 (val=-7.73, SHAP=+1.25, 9.7%) 5. V4 (val=6.54, SHAP=+1.18, 9.2%) Mitigating: V25 (SHAP=-1.15), V8 (SHAP=-0.83)

Automated Testing & Reproducibility (66 / 66 Passing)

Loader & Data: 24 Tests Passing	Feature Pipeline: 35 Tests Passing	Model Inference: 3 Tests Passing	SHAP & Risk Engine: 4 Tests Passing
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GitHub Repository & Delivery

- Repository: uc-believes/Credit-Card-FRAUD_ML
- Branch: main · Commits: 509efc3, 041f6c3
- Artifacts: Models, pipelines, reports & plots stored.

Phase 4 Next Steps

- Web Dashboard: Build modern Flask UI for analysts.
- Investigation Queue: Interactive transaction triage.
- Interactive SHAP View: Real-time waterfall charts.